

can also communicate with other daemons to manage Docker services.

The Docker client (also called 'docker'), with which many Docker users interact with Docker, can communicate with more than one daemon.

A Docker registry stores Docker images. Docker Hub and Docker Cloud are public registries that anybody can use, and Docker is configured to look for images on Docker Hub, by default.

Note: When we use the `docker pull` or `docker run` commands, the required images are pulled from the configured registry. When we use the `docker push` command, the image is pushed to our configured registry.

When we use Docker, we are creating and using images, containers, networks, volumes, plugins and other various such objects. These are called Docker objects.

Thus containers are fundamentally changing the way we develop, distribute and run software on a daily basis.

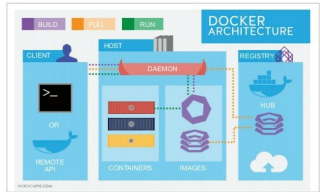


Figure 1: Docker container architecture (<http://apacheboost.com/kh/what-is-a-docker-container-for-beginners/docker-architecture/>)

These developments and advantages of containers help in the advancement of microservices technology. Microservices are small services running as separate processes, where each service is lined up with separate business capabilities. When one lists the advantages of microservices over monolithic

Table 1: A brief history of containers

Year	Container/Technology	Details	Remarks
1979	UNIX V7 (chroot)	The 'chroot' system call was invented to change the root directory of a process and its children to a new FS location.	To provide isolated hardware space
2000	Free BSD Jails	Partitions the system into smaller independent systems called Jails.	Assigns the IP address and is custom installable
2001	Linux Vservers	Securely partitions resources on a computer file system by patching the Linux kernel.	
2004	Oracle Solaris containers	A zone is a virtualised operating system environment created within a single instance of the Solaris operating system. These zones act as complete isolated servers (this isolation prevents processes that are running in one zone from monitoring or affecting processes that are running in other zones).	Zones act as completely isolated servers
2005	OpenVZ (OpenVirtuzzo)	Similar to Solaris containers and renamed as control groups.	
2006	Process containers	By Google, control groups.	
2008	LXC (Linux containers)	Developed by cgroups and namespaces.	Work on the vanilla Linux kernel
2011	Warden	Developed by Cloud Foundry and works on any Linux based OS.	Isolation and daemon running
2013	LMCTFY (Let me contain that for you)	Google's container stack.	This was stopped in 2015
2013	Docker	Developed by Solomon Hykes. Most efficient container management system ever. Uses own library (lib container).	Earlier this was known as 'dotcloud'
2016	Windows containers	Under development.	